Francis Okyere

Ph.D. Student in Statistics
Florida State University

Email: kokyere.gh@gmail.com Phone: (850) 300-9944

Website — Google Scholar — ResearchGate

Research Profile

Statistician specializing in high-dimensional statistics, interpretable machine learning, statistical learning, regularization methods, and predictive modeling with applications in healthcare, education, biomedical research, and public policy.

Research Interests

- High-dimensional statistics
- Statistical learning
- Penalized regression
- Feature selection
- Explainable artificial intelligence
- Healthcare analytics
- Educational data science
- Predictive modeling
- Time series analysis

Education

Florida State University
Ph.D. in Statistics (In Progress)

Florida State University
M.S. Statistics (Applied Statistics), 2026

University for Development Studies, Ghana
B.Sc. Mathematical Sciences (Statistics), 2012

Academic Appointments

Graduate Teaching Assistant, Department of Statistics
Florida State University 2024–Present

Instructor
Wisconsin International University College, Ghana 2023–2024

Senior Research Officer
Ghana TVET Service 2018–2022

Teaching Experience

- STA 2023 – Fundamental Business Statistics
- STA 2122 – Introduction to Applied Statistics
- STA 2171 – Statistics for Biology

Peer-Reviewed Publications

1. Okyere, F., & Sherif, M. (2026). Simulation-Enhanced Regression Analysis of Attendance and Prior Achievement as Predictors of Student Academic Outcomes. *Florida Journal of Educational Research*.
2. Okyere, F., Mahama, F., Yemidi, S., & Krampa, E. K. (2015). An Econometric Model for Inflation Rates in the Volta Region of Ghana. *IOSR Journal of Economics and Finance*.
3. Okyere, F., & Kyei, L. (2014). Behavioral Pattern of Ordinance Marriages in Accra Metropolis: An Application of Multiplicative SARIMA Model. *International Journal of Modern Mathematical Sciences*.
4. Okyere, F., & Kyei, L. (2014). Temporal Modelling of Producer Price Inflation Rates of Ghana. *IOSR Journal of Mathematics*.
5. Okyere, F., & Nanga, S. (2014). Modelling the Pattern of Monetary Policy Rates of Ghana. *IOSR Journal of Mathematics*.
6. Okyere, F., & Nanga, S. (2014). Forecasting Weekly Auction of the 91-Day Treasury Bill Rates of Ghana. *IOSR Journal of Economics and Finance*.
7. Okyere, F., & Mensah, C. (2014). Empirical Modelling and Model Selection for Forecasting Monthly Inflation of Ghana. *Mathematical Theory and Modeling*.

Manuscripts Under Review

1. Okyere, F. (2026). An Interpretable Statistical Learning Framework for Binary Classification: An Application to Student Stress Prediction. Submitted to *Computational Statistics*.
2. Okyere, F., Amuyao, F. M., & Mohammed, S. (2026). Interpretable Forecasting of FIFA World Cup Tournament Progression Using Penalized Logistic Regression and Bootstrap Stability Selection. Submitted to *Quality & Quantity*.
3. Mohammed, S., Kyei, L., Okyere, F., & Amuyao, F. (2026). Assessing Latent Heterogeneity in Breast Cancer Survival: A Frailty Model Approach Using METABRIC Data. Submitted to *BMC Medical Research Methodology*.

4. Okyere, F., & Nyanney, M. (2025). Stability and Interpretability of Penalized Logistic Regression Models for Breast Cancer Risk Prediction. Submitted to *PLOS ONE*.

Current Research Projects

- Stability in high-dimensional statistical models
- Bootstrap stability selection
- Interpretable statistical learning
- Risk prediction in healthcare
- Sports analytics
- Educational data science

Training and Professional Certifications

1. Teaching Statistics Thinking at Curry College (2026)
2. Human Subjects Research (Biomedical/Clinical) (2026)
3. Human Subjects Research (Social/Behavioral) (2026)
4. Research Security Training (Combined) (2026)

Technical Skills

Programming: R, SAS, Excel

Methods: LASSO, Ridge, Elastic Net, Logistic Regression, Random Forest, Bootstrap, Simulation, Time Series, GARCH

Professional Memberships

American Statistical Association

Professional Service

Peer Reviewer, Florida Journal of Educational Research

References

Available upon request.